

Quantifying uncertainty: A comparison of Bayesian and frequentist meta-analytic frameworks applied to lead exposure and children's IQ

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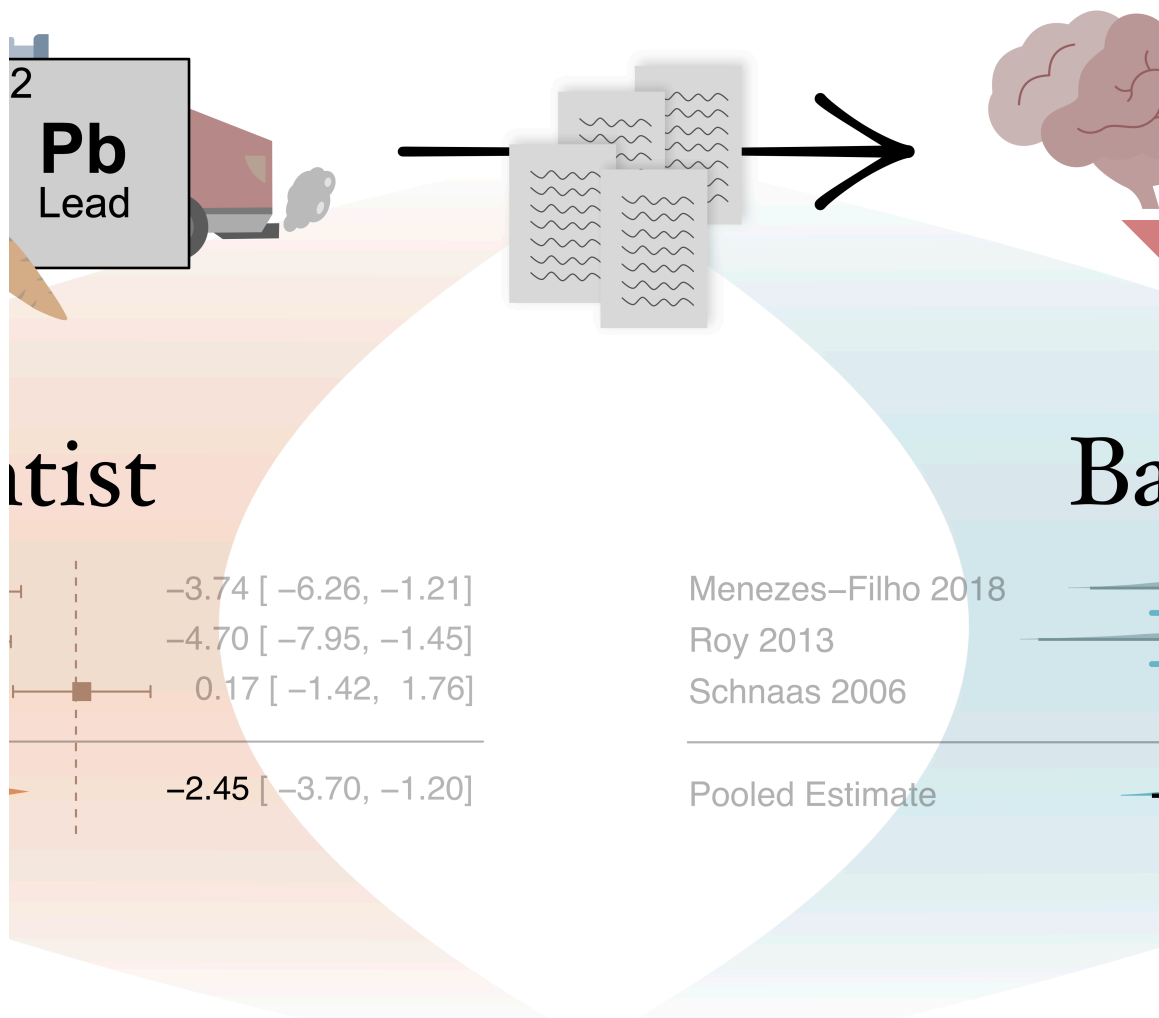
Uncertainty quantification

Lead exposure

Environmental epidemiology

Graphical abstract

Comparing meta-analytic framework Lead exposure and children's IQ



Pooled estimates are roughly equal



Abstract

Background: Uncertainty quantification is central to evidence synthesis for risk assessment and policy-making, but frequentist meta-analytic approaches - currently predominant in epidemiology - often fall short in this regard. Bayesian meta-analysis provides probabilistic uncertainty quantification. To evaluate differences in uncertainty quantification and applicability for risk assessment, we compared frequentist and Bayesian frameworks, using lead exposure and children's IQ as a case study.

Methods: We systematically searched PubMed and Scopus for epidemiological studies on blood lead level (BLL) and IQ score in children. After screening, risk of bias (RoB) in the identified studies was assessed and regression coefficients were harmonized to the $\log_e$ scale. In both meta-analyses, we included the same effect estimates, assumed a multilevel structure and used random-effects models. The Bayesian meta-analysis employed a hierarchical model with weakly-informative priors. Sensitivity analyses examined robustness to RoB and to prior specification. τ^2 and I^2 were estimated as measures of between-study heterogeneity.

Results: Twelve studies were included in the meta-analysis. The frequentist framework yielded a pooled estimate of -2.45 (95% confidence interval: -3.7 to -1.2) IQ points per unit increase in $\log_e$ -transformed BLL ($\mu\text{g/dL}$). The Bayesian model yielded a mean pooled estimate of -2.28 (95% credible interval: -3.56 to -1.06). Between-study heterogeneity was substantial (frequentist: $\tau^2 = 3.3$, $I^2 = 85.0\%$; Bayesian: posterior median $\tau^2 = 3.3$, $I^2 = 84.7\%$). The Bayesian approach provided a full posterior distribution for heterogeneity, directly quantifying uncertainty from between-study variation. Sensitivity analyses showed both frameworks were robust to RoB, and Bayesian estimates were insensitive to tested priors. The Bayesian approach was less sensitive to outlier studies reporting substantial uncertainty, resulting in a lower pooled effect compared to the frequentist approach. This regularisation is particularly valuable in small meta-analyses, where frequentist heterogeneity estimates are unstable — though it relies on heterogeneity priors that are carefully grounded in domain knowledge.

Conclusions: We provide updated pooled estimates for the effect of lead exposure on children's IQ. The Bayesian framework provides posterior distributions, enabling direct probability statements, e.g. about exceeding risk thresholds, and comparative assessments between pollutants. These features are particularly valuable for risk assessment and policy-making.

Background

Lead (Pb) is a pervasive environmental pollutant originating from anthropogenic and natural sources. Although concentrations have declined globally in past decades following the phase-out of leaded gasoline, lead remains present in the environment and is detected in human blood samples [1,2]. In Europe, primary exposure sources include dietary products, drinking water, and indoor dust [3,4]. The most recent Global Burden of Disease study estimated that in 2023, 3.4 million Disability-Adjusted Life Years (DALYs, 95% uncertainty interval (UI): 2.7 to 4.0 million) were attributable to lead exposure in the European Union (EU) [5]. This underscores the continued public health significance even of low-level exposure.

Exposure to lead is associated with multiple adverse health outcomes across the human life course, including decreased kidney function [6], liver disease [7], coronary artery atherosclerosis [8], cardiovascular mortality [9], infertility in women [10], attention-deficit/hyperactivity disorder in children [11], and antisocial behavior [12]. Among these endpoints, impaired cognitive development in children — observable as reduced intelligence quotient (IQ) scores — represents a particularly critical endpoint with high public health relevance [13–16]. Children are especially vulnerable to the neurotoxic effects of lead due to their developing nervous systems and relatively high exposure per body mass. Because lead-associated IQ decrements are irreversible and persist throughout life, they lead to substantial and enduring societal costs, making continued efforts to accurately characterize the exposure-response relationship important for evidence-based policy [17].

Blood lead levels (BLL) in the EU have declined markedly and now predominantly fall below 5 µg/dL, well below the concentrations examined in many earlier studies (<10 µg/dL) [18]. Importantly, there is no evidence for a threshold concentration below which the effects of lead on IQ can be assumed to be negligible [19,20]. There is evidence for a supralinear exposure-response relationship with steeper IQ decrements per unit increase in BLL at low concentrations compared to higher concentration ranges [16,21]. These findings highlight the need for an updated quantitative synthesis that reflects both recent epidemiological evidence and the lower exposure range now predominant in European populations, incorporating recent data to inform risk assessment at currently relevant exposure levels.

Existing burden of disease assessments at national and global scales [5,17,22–24] have predominantly relied on the foundational study by Lanphear et al. (2005), a pooled analysis of seven cohort studies, or subsequent re-analyses of the same data [16,25]. While these landmark studies have substantially shaped research and policy on childhood lead exposure, methodological issues and shortcomings have been raised. These include the use of performance IQ instead of full-scale IQ and an erroneous back-transformation of BLL in one cohort in the Lanphear et al. study [25,26]. These limitations, combined with the availability of more recent epidemiological data, underscore the need not only for an updated evidence synthesis, but also for methodologically robust approaches that can provide reliable exposure-response estimates to be used in burden of disease and risk assessments.

Methodological advances in meta-analysis offer an opportunity to enhance the utility of synthesized estimates for burden of disease and risk assessment applications. While frequentist

meta-analysis has been a conventional approach in environmental epidemiology, Bayesian meta-analysis provides a coherent framework for uncertainty quantification through posterior distributions that inherently incorporate both within-study and between-study variability [27]. Unlike frequentist confidence intervals, which are often misinterpreted as having a 95% probability of containing the unknown parameter [28], Bayesian credible intervals directly represent the probability distribution of the parameter of interest, facilitating transparent communication of uncertainty. Furthermore, Bayesian posterior distributions can be directly propagated into probabilistic burden of disease and risk assessments, providing seamless integration with probabilistic frameworks in public health decision-making. By comparing frequentist and Bayesian approaches, we explore how the analytical framework and its approach to uncertainty quantification influence exposure-response estimates and their application.

The aim of this study is twofold: first, to synthesize and quantitatively pool available epidemiological findings on the association between BLL and IQ scores in children; and second, to conduct both frequentist and Bayesian meta-analyses, comparing how these analytical frameworks impact exposure-response estimates and uncertainty quantification.

Methods

Literature Review

This review adheres to the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines [29]. The checklist can be found in additional file 1. The reporting of the Bayesian methods and their results was guided by the Bayesian Analysis Reporting Guidelines (BARG) [30].

We conducted a literature review to identify published studies investigating the association between postnatal (concurrent) exposure to Pb and IQ loss in children. This study was conducted within the EU-funded project “*Partnership for the Assessment of Risks from Chemicals*” (PARC) [31]. The literature review presented here was used in multiple case studies within PARC, including one on the health burden of (methyl)mercury and lead on IQ loss. Here, we will only report results for concurrent lead exposure and IQ loss, but the figures and tables documenting the literature review contain parts of the search for other associations and exposure timeframes. We searched for articles published between January 2005 and July 2023. The year 2005 was chosen because burden of disease studies have predominantly relied on a landmark paper published in that year [16] and an update of the evidence base is therefore warranted, particularly given that lead exposure concentrations have decreased substantially since then.

Search queries in PubMed (MEDLINE) and Scopus included the following terms: substance (Pb - including Medical Subject Headings [MeSH] terms), outcome (IQ and neurodevelopment), population (children) and results (association, exposure-response function [ERF]) (additional file 2).

All titles and abstracts were screened by two reviewers. Any conflicts were resolved by a third person. Full texts were then assessed by one reviewer. Publications were considered relevant according to predefined PECOTS – Population, Exposure, Comparator, Outcome, Timing of exposure and Setting (Table 1).

Table 1: PECOTS criteria

PECOTS domain	Inclusion criteria
Population	Human population < 18 years of age (infants, children, teenagers) OR pregnant women and women in childbearing age
Exposure	Exposure to lead and methylmercury through HBM/internal exposure data (urine, blood, hair, bones).
Comparators	Investigating associations between exposure and outcome for single substance or a mixture AND reporting regression coefficient and statistical uncertainty
Outcomes	IQ shift in children < 18
Timing	Current exposure of the children (post-natal) OR current exposure of pregnant women (pre-natal). Both are considered medium- to long-term timing of exposure.
Settings	All settings (location)

HBM, human biomonitoring

We excluded reviews and background articles that inventoried the literature without quantitatively summarizing the results. We also excluded references written in languages other than English. For this present study, only studies that used a continuous, log-transformed BLL scale were included, so that effect estimates could be harmonized. The exposure timeframe for this study was restricted to concurrent, i.e. IQ and BLL measurements must have been conducted at the same time. Descriptive study data was extracted using a grid based on the one proposed by the Office of Health Assessment and Translation within the US National Toxicology Program [32]. For all included studies, effect estimates from the most-adjusted model were extracted and the adjustment set was recorded.

Risk of Bias Assessment

To assess study quality, we conducted a risk of bias assessment using the rapid Risk of Bias (RoB) tool with a specific focus on human observational epidemiological studies [33]. Each included publication was assessed for RoB in the following domains: selection, exposure, outcome, confounding, follow-up, analysis, and selective reporting. In total, each study was evaluated using 15 items and scored as either high RoB (-), some concerns (+/-), low RoB (+) or no information (?). If an item was not applicable to the study type (e.g. follow-up in cross-sectional studies), they were marked as “n.a.”. The overall RoB for each individual study was then categorized as either low, moderate or high, following two different summary metrics provided by the RoB tool (for more details, please refer to Plate K, Knüppel S, Hornbacher A, Greiner M, Müller-Graf C. [33]). Each publication was first assessed by one researcher. Subsequently, consensus with the large language model (LLM) Claude Sonnet 4.6 (Anthropic) was sought. This was done by uploading the list of all RoB items along with the respective publication. Then, the LLM was prompted to assess the study, using the RoB items. If there were any deviations in the judgement of the researcher and the LLM, the respective item was revisited by the researcher who made the final decision.

Harmonization of Effect Estimates

Studies investigating the association between lead exposure and IQ in children have made different assumptions regarding the shape of the ERF. While some studies assume a linear relationship [34,35], this assumption may only be valid within very narrow concentration ranges [25]. Studies that tested multiple shapes of the ERF have concluded that a log-linear relationship, where BLL are logarithmically transformed, represents a more appropriate statistical model [25,36].

The general form of regression model used for a log-linear association is:

$$y_{pred} = \alpha + \beta \times \log_b(X) + \varepsilon \quad (1)$$

Where y_{pred} represents the IQ score, α is the intercept, β is the regression coefficient (slope), X denotes BLL, b is the logarithmic base, and ε is the error term.

In the literature, different logarithmic bases are used to transform BLL, including the natural logarithm ($\log_e$, where $e \approx 2.718$), base 2, and base 10. To enable pooling of the effect estimates, we harmonized all regression coefficients and their standard error (SE) to increments on the natural logarithm scale ($\log_e$). The calculations are documented in additional file 3. In some publications, BLL was not on a logarithmic but on an untransformed, linear scale. These were excluded from the meta-analysis, as transformation of effect estimates for linear BLL to log-linear could not be justified.

Meta-Analysis

In order to quantitatively synthesize the available epidemiological literature on the association of children’s blood lead levels and IQ, we conducted separate meta-analyses in the Bayesian and frequentist analytical framework, and compared their results. To make the results of both approaches comparable, both included the same effect estimates from the identified publications, and featured analogous model structure. We estimated pooled effect estimates for the loss of IQ points per continuous unit increase in $\log_e$ -transformed BLL ($\mu\text{g/dL}$). R version 4.4.2 (2024-10-31) was used for all computations [37]. All code for this study is publicly available on GitHub [38].

Bayesian Meta-Analysis

In the Bayesian framework, we employed a random-effects meta-analysis model. Each study’s observed effect size $\hat{\theta}_k$ is modelled as a noisy reflection of the true underlying effect in that study’s sample θ_k , with the noise represented by the within-study sampling error σ_k (treated as known based on reported study statistics). These study-specific true effects are themselves modelled as draws from an overarching normal distribution which describes the true average effect across studies μ and the between-study heterogeneity τ .

$$\begin{aligned} \hat{\theta}_k &\sim \mathcal{N}(\theta_k, \sigma_k) \\ \theta_k &\sim \mathcal{N}(\mu, \tau) \end{aligned} \quad (2)$$

Where $\hat{\theta}_k$ is the observed effect size for study k , σ_k is the within-study sampling standard deviation for study k , θ_k is the true underlying effect for study k , μ is the overall pooled effect and τ is the between-study standard deviation (heterogeneity). The Bayesian model was fit using

the R package `brms` version 2.22.0 [39]. Sampling was done using Markov chain Monte Carlo (MCMC). We ran four chains with 4,000 iterations each, of which 2,000 iterations were discarded as warmup, resulting in 8,000 total post-warmup draws. MCMC convergence was assessed using traceplots and the Gelman-Rubin statistic ($\hat{R}$). Chain resolution was assessed using the effective sample size (ESS). Heterogeneity τ was directly modelled as a parameter (Equation 2). I^2 was calculated using the typical within-study variance calculated with `metafor` for each τ^2 posterior sample. Because the posterior distribution of τ^2 is right-skewed, we summarise τ^2 and I^2 by their posterior medians rather than their posterior means, as the median represents the typical value more faithfully and keeps the reported point estimates consistent with their credible intervals (for further detail, see additional file 4).

In the Bayesian framework, domain knowledge is elicited and captured as probability distributions (priors) which are subsequently updated by the newly observed data, resulting in a posterior distribution. Prior distributions are specified for all model parameters, i.e. the overall pooled effect estimate μ and between-study standard deviation (heterogeneity) τ . For both parameters, weakly-informative priors were chosen to softly constrain the posterior distributions to realistic ranges of results. The priors are specified on the same scale as the harmonized effect estimates (IQ shift per unit increase in $\log_e$ -transformed BLL).

For the overall pooled effect μ , a normally distributed prior centered at -1 with a standard deviation (SD) of 2 was chosen. This reflects the prior belief that the overall pooled effect lies between -5 and 3 with $\sim 95\%$ probability. The center point of the prior was chosen to be -1 because the negative effects of lead on the developing nervous system and its toxicological mechanisms of action are well established [19]. Since variance terms, such as the between study heterogeneity τ , must be positive, a truncated normal distribution centered at 0 with a SD of 2 was specified. This reflects the prior belief that heterogeneity τ is small to moderate, but does not exclude large heterogeneity if this is present in the included studies. In summary, the following priors were specified for the Bayesian model:

$$\begin{aligned}\mu &\sim \mathcal{N}(-1, 2) \\ \tau &\sim \mathcal{N}^+(0, 2)\end{aligned}\tag{3}$$

Prior predictive checks were conducted using the `bayesplot` package [40] to verify that the specified priors were appropriate. Posterior predictive checks were carried out to assess if the model structure is able to capture the underlying data well. Additionally, a sensitivity analysis with alternative priors was conducted, as described below.

Frequentist Meta-Analysis

For the frequentist meta-analysis, we employed a random-effects model, assuming the same multilevel structure as for the Bayesian model (Equation 2). Parameters were estimated via the restricted maximum-likelihood estimation from the R package `metafor` version 4.8-0 [41]. The I^2 statistic was calculated as a measure of heterogeneity, representing the proportion of total variation attributable to between-study heterogeneity τ^2 .

Sensitivity Analysis

We conducted two sets of sensitivity analyses. To examine the sensitivity of the results to possibly biased study estimates, we ran both the frequentist and Bayesian models again twice, excluding (1) studies with a high RoB rating and (2) with a high or moderate RoB.

To examine the sensitivity of the results to the choice of prior distributions, we fit the Bayesian model again for two alternative sets of priors that we consider to be extreme but still reasonable. The first set of priors was more informative (narrower) than the main priors, with $\mu \sim \mathcal{N}(-1, 1)$ and $\tau \sim \mathcal{N}^+(0, 1)$. The second set of priors was less informative (wider) with $\mu \sim \mathcal{N}(0, 6)$ and $\tau \sim \mathcal{N}^+(0, 6)$. Note that for the main effect, the broader prior is centered around 0, giving equal weight to the probability of a protective and detrimental effect of lead.

Results

Literature Review

The systematic review was conducted with a larger scope and was used as a basis for multiple case studies within the PARC project, which is why it includes another substance (mercury, Hg). The systematic search in PubMed and Scopus yielded 1,073 records for lead. After duplicate removal, title and abstract of 974 records (Pb & Hg) was screened, which left 149 records (Pb & Hg) for full text screening. After this last screening, 36 records remained for Pb. These were then assessed for their eligibility for the meta-analysis. Twelve studies were excluded because they either transformed the BLL variable into categories or left it on a linear (untransformed) scale. Four studies were excluded because they exclusively assessed the effects of prenatal lead exposure. Furthermore, four studies were excluded because the cohorts overlapped with another study and would lead to duplicates. Four additional studies were excluded because they either solely reported effects of chemical mixtures, used non-linear models (including the use of splines) or failed to report essential statistics. Finally, twelve studies were included in the meta-analysis. Study characteristics and regression coefficients (effect estimates) are shown in Table 2.

Table 2: Study Characteristics

First author, year	Study population (N)	Geographical area	Mean age / age range	BLL indicator	BLL value ($\mu\text{g/dL}$)	IQ measure	Mean IQ	Adjustment set (confounder)	Original regression coefficient	Harmonized regression coefficient	Harmonized SE
Bauer 2020	635	Province of Brescia, Italy	10-14	Median	1.3	WISC-III (FSIQ)	107.1	Sex, age, SES, hemoglobin, HOME score and HOME score squared	Log(e) -0.10	0.87	-0.10 0.87
Crump 2013	1355	USA (Boston, Cleveland, Cincinnati, Rochester), Mexico, Australia (Port Pirie) and Yugoslavia	5-7	Mean	10.0	WISC (FSIQ)	94.4	Site, HOME score, birth weight, maternal IQ, maternal education, maternal alcohol use, maternal	Log(e) -3.31	0.63	-3.31 0.63

ADHD-RS, Attention Deficit Hyperactivity Disorder Rating Scale; BMI, Body Mass Index; CPT, Continuous Performance Test; ERF, exposure-response function; FSIQ, full scale IQ; HOME, Home Observation for Measurement of the Environment; SE, standard error; SES, socioeconomic status; WASI, Wechsler Abbreviated Scale of Intelligence, WISC, Wechsler Intelligence Scale for Children; WPPSI, Wechsler Preschool and Primary Scale of Intelligence

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First author, year	Study population (N)	Geographical area	Mean age / age range	BLL indicator	BLL value ($\mu\text{g}/\text{dL}$)		IQ measure	Adjustment set (founder)	Original regression coefficient	Harmonized regression coefficient		Harmonized SE
					un-exposed	IQ measure				Original SE	Harmonized SE	
								tobacco use and birth order				
								Care-giver FSIQ (WASI-2), BMI at age 12, and a measure of neighborhood-level deprivation based on census-tract level socioeconomic indicators				
Dantzer 2020	224	USA (Greater Cincinnati, Ohio metropolitan region)	12	Mean	0.6	WISC-IV (FSIQ)	101.6	Log(10)	3.15	3.71	-5.71	1.61

ADHD-RS, Attention Deficit Hyperactivity Disorder Rating Scale; BMI, Body Mass Index; CPT, Continuous Performance Test; ERF, exposure-response function; FSIQ, full scale IQ; HOME, Home Observation for Measurement of the Environment; SE, standard error; SES, socioeconomic status; WASI, Wechsler Abbreviated Scale of Intelligence, WISC, Wechsler Intelligence Scale for Children; WPPSI, Wechsler Preschool and Primary Scale of Intelligence

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					unless state-wise	IQ measure - instrument				Original SE	Harmonized SE	
Desrochers-Couture 2018	606	Canada (10 different study sites)	3-4	Mean	0.7	WPPSI-III (FSIQ)	107.0	Child age, child sex, evaluation site, marital status, familial income, HOME score, Parenting Stress Index, and cord blood lead (log-2 scale)	Log(2) 0.01	0.04	0.02	0.06
Earl 2016	127	Australia, New South Wales, Port Pirie and Bro-	7-8	Mean	3.9	WISC-IV (FSIQ)	100.5	Current maternal smoking,	Log(10) 3.50	5.10	-5.86	2.22

ADHD-RS, Attention Deficit Hyperactivity Disorder Rating Scale; BMI, Body Mass Index; CPT, Continuous Performance Test; ERF, exposure-response function; FSIQ, full scale IQ; HOME, Home Observation for Measurement of the Environment; SE, standard error; SES, socioeconomic status; WASI, Wechsler Abbreviated Scale of Intelligence, WISC, Wechsler Intelligence Scale for Children; WPPSI, Wechsler Preschool and Primary Scale of Intelligence

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		Ken Hill					breast-feeding, income, HOME score and stressful life events			
Hong 2015	839	South Korea (two urban cities, two industrial cities, and one rural district)	8-11	Mean	1.8	Korean WISC.1 (FSIQ)	Demographic variables and ADHD-RS and CPT scores, plus log10-transformed environmental chemical concentrations (blood mercury and manganese)	Log(10) 7.23	3.14	-3.14 1.37

ADHD-RS, Attention Deficit Hyperactivity Disorder Rating Scale; BMI, Body Mass Index; CPT, Continuous Performance Test; ERF, exposure-response function; FSIQ, full scale IQ; HOME, Home Observation for Measurement of the Environment; SE, standard error; SES, socioeconomic status; WASI, Wechsler Abbreviated Scale of Intelligence, WISC, Wechsler Intelligence Scale for Children; WPPSI, Wechsler Preschool and Primary Scale of Intelligence

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					un- less state- dother- wise)	IQ mea- sure- ment					Original	co-efficient	
									concentrations, and creatinine-standardized urine concentrations of cotinine, phthalate metabolites, and Bisphenol A)				
Lee 2021	1502	South Korea (Seoul, Gyeonggi, 6 and Incheon provinces)	Mean	1.4		Korean WISC (FSIQ)	100.9		Maternal education, exposure to en-	Log(e) -3.83	2.25	-3.83	2.25

ADHD-RS, Attention Deficit Hyperactivity Disorder Rating Scale; BMI, Body Mass Index; CPT, Continuous Performance Test; ERF, exposure-response function; FSIQ, full scale IQ; HOME, Home Observation for Measurement of the Environment; SE, standard error; SES, socioeconomic status; WASI, Wechsler Abbreviated Scale of Intelligence, WISC, Wechsler Intelligence Scale for Children; WPPSI, Wechsler Preschool and Primary Scale of Intelligence

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					unless state- dother- wise)	IQ mea- sure- ment				Original	co- efficient	
								vi- ronmen- tal tobacco smoke dur- ing preg- nancy, ma- ter- nal age and ma- ter- nal IQ				
Lucchini 2019	299	South- ern Italy (City of Taranto, Puglia region, 5 sub- ar- eas, in- cremen- tal distance	6-12	Me- dian	8.4 ng/ mL	WISC- IV (FSIQ)	106.0	Age, sex, ma- ter- nal, Stan- dard Progres- sive Matrice, non- ver- bal in- tel-	Log(e) per ng/ mL -1.10	0.84	-1.10	0.84

ADHD-RS, Attention Deficit Hyperactivity Disorder Rating Scale; BMI, Body Mass Index; CPT, Continuous Performance Test; ERF, exposure-response function; FSIQ, full scale IQ; HOME, Home Observation for Measurement of the Environment; SE, standard error; SES, socioeconomic status; WASI, Wechsler Abbreviated Scale of Intelligence, WISC, Wechsler Intelligence Scale for Children; WPPSI, Wechsler Preschool and Primary Scale of Intelligence

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							lec-tual efficiency, HOME score and distance from the point source			
Lucchini 2012	299	Northern Italy (Valcamonica and Garda Lake regions)	11-14	Median	1.5	WISC-III (FSIQ)	106.3	Log(e)-3.48	1.48	-3.48 1.48
Menezes-Filho 2018	155	Brazil (Municipality of Sinop, Mato Grosso do Sul)	7-12	Mean	1.6	WASI (FSIQ)	75.5	Log(10)-8.61	2.97	-3.74 1.29

ADHD-RS, Attention Deficit Hyperactivity Disorder Rating Scale; BMI, Body Mass Index; CPT, Continuous Performance Test; ERF, exposure-response function; FSIQ, full scale IQ; HOME, Home Observation for Measurement of the Environment; SE, standard error; SES, socioeconomic status; WASI, Wechsler Abbreviated Scale of Intelligence, WISC, Wechsler Intelligence Scale for Children; WPPSI, Wechsler Preschool and Primary Scale of Intelligence

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		itan region of Salvador, Bahia)								
Roy 2013	708	India (Chennai)	3-7	Mean	11.5	Bi-net-Kamath Scale of Intelligence (Tamil version of Stanford-Bi-net) (FSIQ)	Age, age ² in months, sex, mid-arm circumference, average monthly household income, parental education, family size	Log(e) -4.70	1.66	-4.70 1.66
Schnaas 2006	2606	Mexico (Mexico City)	6-10	Mean	6.7	WISC-Revised (FSIQ)	Maternal IQ, child	Log(e) 0.17	0.81	0.17 0.81

ADHD-RS, Attention Deficit Hyperactivity Disorder Rating Scale; BMI, Body Mass Index; CPT, Continuous Performance Test; ERF, exposure-response function; FSIQ, full scale IQ; HOME, Home Observation for Measurement of the Environment; SE, standard error; SES, socioeconomic status; WASI, Wechsler Abbreviated Scale of Intelligence, WISC, Wechsler Intelligence Scale for Children; WPPSI, Wechsler Preschool and Primary Scale of Intelligence

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First author, year	Study population (N)	Geographical area	Mean age / age range	BLL indicator	BLL value ($\mu\text{g}/\text{dL}$)		IQ measure - Mean IQ	Adjustment set (founder)	Original regression coefficient	Harmonized regression coefficient		Harmonized SE
					unless state-wise)	IQ measure - instrument				Original SE	Harmonized SE	
								sex, birth weight, socioeconomic status (based on parental education, occupation, and income), indicator for first FSIQ testing				

ADHD-RS, Attention Deficit Hyperactivity Disorder Rating Scale; BMI, Body Mass Index; CPT, Continuous Performance Test; ERF, exposure-response function; FSIQ, full scale IQ; HOME, Home Observation for Measurement of the Environment; SE, standard error; SES, socioeconomic status; WASI, Wechsler Abbreviated Scale of Intelligence, WISC, Wechsler Intelligence Scale for Children; WPPSI, Wechsler Preschool and Primary Scale of Intelligence

A PRISMA flow diagram summarizing the review process is shown in Fig. 1 [29]. Detailed information on the screening process and exclusion reasons can be found in additional file 3.

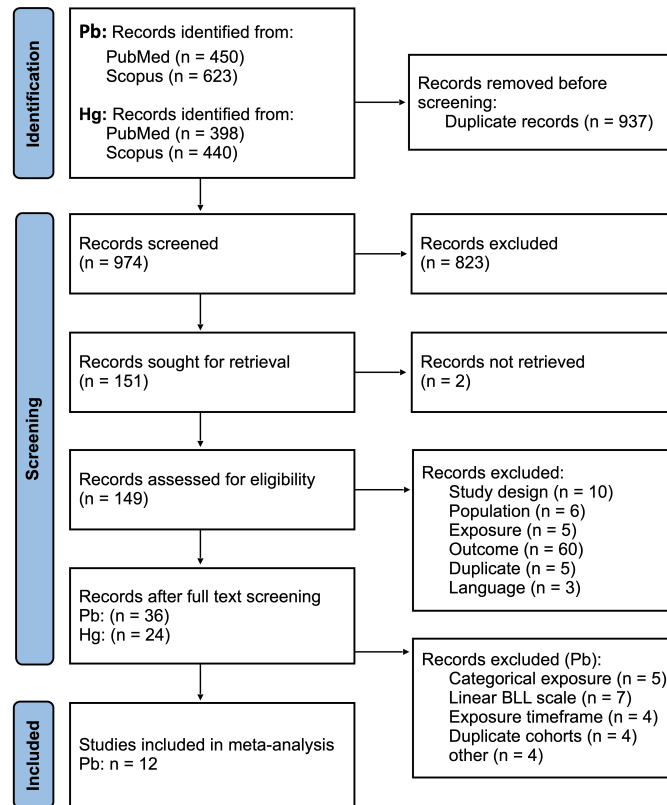


Figure 1: Fig. 1: PRISMA flow diagram. BLL, blood lead level; Pb, lead; Hg, mercury

The size of the study population in included studies ranged from 127 [42] to 1,355 [25]. Study designs included cross-sectional [36,43–47] and cohort studies [25,42,48–50]. The included studies were conducted in multiple different regions of the world (North America: 4, South America: 1, Europe: 3, Asia: 3, Australia: 1) and some populations were sampled in hotspot regions with presumably higher lead concentrations [42,43,51]. The study by Roy A, Ettinger AS, Hu H, Bellinger D, Schwartz J, Modali R, et al. [47] reported implausibly large effect sizes. We contacted the corresponding author, who confirmed that decimal points were missing in the publication.

Risk of Bias

The rapid RoB assessment resulted in eight studies with a low RoB rating [36,43,44,47–51], two with a moderate RoB [45,46] and two studies with a high RoB rating [25,42]. Among the twelve included studies, items with a high overall RoB were *non-response rate* (7x high RoB, 5x some concerns), where some studies did not report the non-response rate or did not discuss whether non-responding individuals might systematically differ from included participants, leading to potential selection bias. Another item with overall high RoB was *accounting for confounding* (2x high RoB, 5x some concerns), where important confounders like maternal IQ were not included or the authors used a data-driven selection approach for covariate selection, when a theory-driven approach was preferable [52]. Finally, in the item *handling of missing values*, some of the studies contain no information on how missing data was handled, while in other studies they

were handled inadequately (3x high RoB, 4x some concerns). Results from the RoB assessment are presented in Table 3.

Table 3: Risk of Bias Assessment

Study, year	Selection			Exposure		Outcome		Con- founding		Follow- up		Analysis		Selec- tive Re- port- ing	Over- all Risk of Bias	
	Eligibility criteria	Comparability of exposure and comparison groups	Non-response rate	Recruitment frame	Accuracy, validity and reliability of exposure measurement	Misclassification of exposure	Accuracy, validity and reliability of outcome measurement	Misclassification of outcome	Accuracy, validity and reliability of confounding measurement	Accounting for confounding	Adequacy of length of follow-up	Amount and proportion of loss to follow-up	Appropriateness of analysis methods	Handling of missing data	Selective reporting of outcomes	Assessment of overall bias
Bauer 2020	+	+/-	-	+	?	+	+	+	+/-	+	n.a.	n.a.	+	+	-	low
Crump 2013	+	+	+/-	-	+/-	-	+	+	-	-	+	+	-	-	+	high
Dantzer 2020	-	+	-	+	+	+	+	+	+	+	+	+	+	+/-	+	low
Desrochers-Couture 2018	+	+	-	+	?	+	+	+	+	+	+/-	+	+	+	+	low
Earl 2016	+	+/-	-	+/-	-	+	+	+	-	+/-	+	?	+/-	-	+/-	high
Hong 2015	+	+	+/-	+	+	+/-	+	+	+/-	+	n.a.	n.a.	+	+	+	low
Lee 2021	+	+	+/-	+	+	+	+	+	+	+	+	+/-	+	+/-	+	low

RoB, risk of bias; (-), high RoB; (+/-), some concerns / moderate; (+), low RoB; (?), no information; (n.a.), not applicable

Table 3: Risk of Bias Assessment

Study, year	Selection				Exposure		Outcome		Con- founding		Follow- up		Analysis		Se- lec- tive Re- port- ing	Over- all Risk of Bias
	El- igi- bil- ity cri- teria	ex- pos- ure and com- par- is- on	Non- re- spon- se rate	Re- cruit- ment time frame	Ac- cu- racy, va- lid- ity and re- lia- bil- ity of ex- pos- ure mea- sure- ments	Mis- clas- si- fi- ca- tion of ex- pos- ure mea- sure- ments	Ac- cu- racy, va- lid- ity and re- lia- bil- ity of out- come mea- sure- ments	Mis- clas- si- fi- ca- tion of out- come mea- sure- ments	Ac- cu- racy, va- lid- ity and re- lia- bil- ity of con- found- ing meas- ure- ments	Ad- equa- cy of follow- up	Amount and pro- pri- ate meth- ods	Ap- pri- ate data anal- ysis of loss to fol- low- up	Se- lec- tive re- port- ing of out- come mea- sure- ments	As- ses- sors of over- all bias		
Luc- chini 2019	+	+	-	-	+	+	+	+	+	+	n.a.	n.a.	+	+/-	+	mod- er- ate
Luc- chini 2012	+	+	+/-	-	+	+	+	+	+/-	+/-	n.a.	n.a.	+	-	+	mod- er- ate
Menezes- Filho 2018	+	+	-	+	+	+	+	+	+/-	+	n.a.	n.a.	+	+/-	+	low
Roy 2013	+	+	-	+	+	+	+	+	+/-	+	n.a.	n.a.	+	+	+	low
Schnaas 2006	+	+	+/-	+	+	+/-	+	+	+	+	n.a.	n.a.	+	+	+	low

RoB, risk of bias; (-), high RoB; (+/-), some concerns / moderate; (+), low RoB; (?), no information; (n.a.), not applicable

Meta Analysis

The prior predictive check indicates that the specified priors are appropriate (additional file 5a). The posterior predictive check showed that the model correctly captures important patterns in the data (additional file 5b). $\hat{R}$ values indicate that all MCMC chains converged (1.00 for both μ and τ). MCMC trace plots showed well-mixed chains and convergence (additional file 5c). ESS indicate that sampling was efficient and central point estimates and credible intervals can be interpreted

with confidence (ESS bulk for μ : 1,796; ESS tail: 2,504, corresponding to 22.5% and 31.3% of total samples respectively).

The Bayesian random-effects meta-analysis yielded a posterior mean pooled estimate of -2.28 (95% credible interval (CrI): -3.56 to -1.06) IQ points per log-unit increase in BLL ($\mu\text{g/dL}$) (see Fig. 2). The frequentist approach produced a comparable pooled estimate of -2.45 (95% confidence interval (CI): -3.7 to -1.2). Substantial between-study heterogeneity was observed under both frameworks. In the Bayesian framework, between-study heterogeneity was estimated as a posterior median τ^2 of 3.3 (95% CrI: 1.0 to 9.5), corresponding to an I^2 of 84.7% (95% CrI: 63.9% to 94.2%). The frequentist model estimated a closely comparable $\tau^2 = 3.3$ (95% CI: 1.0 to 11.7), with $I^2 = 85.0\%$ (95% CI: 62.7% to 95.2%). Additional file 6 contains the full model code and output for both frameworks. The results of both frameworks are shown as forest plots in Figure 2.

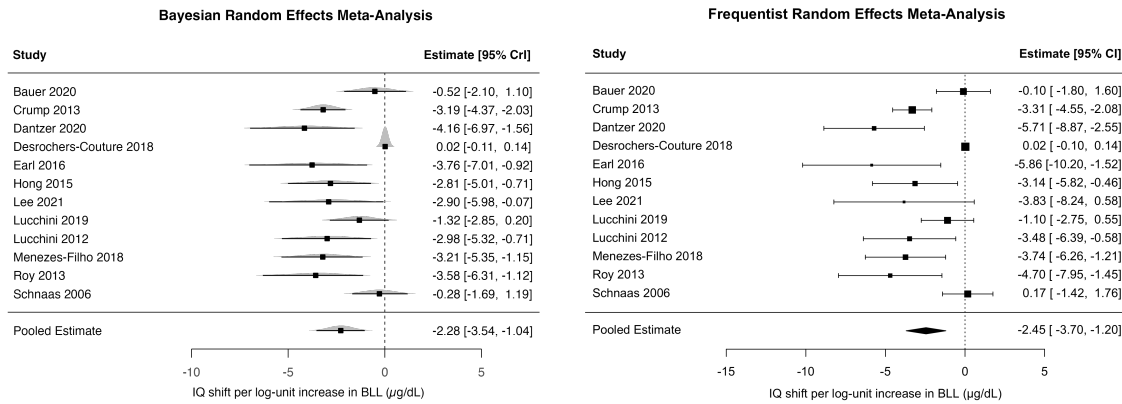


Figure 2: Fig. 2: Forest plots for Bayesian (left) and frequentist (right) meta-analyses. BLL, blood lead level; CI, confidence interval; CrI, credible interval

The single study estimates shown in Fig. 2 differ between frameworks because Bayesian shrinkage pulls individual study estimates toward the pooled mean, borrowing information from the overall mean. This has no equivalent in the frequentist framework, where the contribution of each study to the pooled effect is determined by inverse-variance weighting. The degree of shrinkage does, however, depend on each study's precision in a way that parallels inverse-variance weighting: more precise studies are shrunk less (and are also weighted more heavily toward the frequentist pooled estimate), while less precise studies are shrunk more (and weighted less).

Sensitivity Analysis

We conducted two sets of sensitivity analyses; results are shown in Table 4. First, in order to test the sensitivity of the results to possibly biased study estimates, we excluded (1) studies with high RoB rating and (2) with high or moderate RoB rating. For both the frequentist and Bayesian frameworks, the strength of the mean pooled effect decreased after removing studies with high risk of bias and slightly increased again after removing high and moderate RoB studies. These shifts in both directions and their small size show that the results are robust to possible individual-study bias.

The second set of sensitivity analyses examined the sensitivity of the results of the Bayesian meta-analysis to more extreme prior settings that can still be considered reasonable. With more informative (narrower) priors ($\mu \sim \mathcal{N}(-1, 1)$, $\tau \sim \mathcal{N}^+(0, 1)$), the posterior mean of μ decreases. This is expected, because the narrow prior restricts the posterior distribution to lower effect ranges.

When using less informative (wider) priors ($\mu \sim \mathcal{N}(0, 6)$, $\tau \sim \mathcal{N}^+(0, 6)$), the posterior mean of μ increases slightly compared to the main model. This set of priors gives more weight to studies reporting high effects with large uncertainty ranges. Additional file 5d shows prior, observed data and posterior distributions of all three prior specifications. This shows that the results are robust to the alternative priors tested in the sensitivity analysis, because the shifts in the pooled estimate are not substantial and do not change the conclusion of our analysis.

Table 4: Results of sensitivity analyses by framework

Framework	Variant	Estimate (95% UI)
RoB		
Bayesian	Full sample	-2.31 (-3.63, -1.08)
	Low RoB only	-1.94 (-3.59, -0.39)
	Low/Medium RoB	-1.94 (-3.32, -0.67)
Frequentist	Full sample	-2.45 (-3.70, -1.20)
	Low RoB only	-2.23 (-3.91, -0.55)
	Low/Medium RoB	-2.14 (-3.50, -0.78)
Priors		
Bayesian	Main	-2.31 (-3.63, -1.08)
	Narrow priors	-1.98 (-2.94, -1.03)
	Wide priors	-2.46 (-4.03, -1.11)
Effect prior		
Bayesian	Skeptical: Normal(-0.33, 2)	-2.22 (-3.50, -1.00)
	Default: Normal(-1, 2)	-2.31 (-3.63, -1.08)
	Enthusiastic: Normal(-3, 2)	-2.48 (-3.84, -1.24)
Heterogeneity prior		
Bayesian	Strong shrinkage: Half-normal(0.67)	-2.25 (-3.41, -1.21)
	Default: Half-normal(2)	-2.43 (-3.90, -1.16)

RoB, risk of bias; UI, uncertainty interval (credible interval for Bayesian and confidence interval for frequentist model results). Effect prior analyses hold the heterogeneity prior at Half-normal(2); heterogeneity prior analyses hold the effect prior at Normal(0, 100), which is effectively flat. Half-normal(s) denotes a Normal(0, s) distribution truncated at zero.

Table 4: Results of sensitivity analyses by framework

Framework	Variant	Estimate (95% UI)
	Weak shrinkage: Half-normal(6)	-2.46 (-3.98, -1.10)

RoB, risk of bias; UI, uncertainty interval (credible interval for Bayesian and confidence interval for frequentist model results). Effect prior analyses hold the heterogeneity prior at Half-normal(2); heterogeneity prior analyses hold the effect prior at Normal(0, 100), which is effectively flat. Half-normal(s) denotes a Normal(0, s) distribution truncated at zero.

Discussion

In this study, we provide an updated pooled effect estimate for the association between children’s lead exposure and IQ, and compare a conventional, frequentist meta-analysis to a Bayesian meta-analysis – a framework that is less used in the field of environmental epidemiology to date.

While previous publications have compared both frameworks, they focus either on evidence-based medicine [53–56] or network meta-analysis [57,58] and thus address methodological challenges that only partially overlap with those encountered in environmental epidemiology. To our knowledge, this is the first study to compare Bayesian and frequentist meta-analytic approaches in this field.

Pooled effect estimates and measures of heterogeneity were largely consistent across both frameworks, though the Bayesian model yielded slightly lower mean pooled effects. This is attributable to both the prior for the overall effect (μ) and the posterior shrinkage of individual study estimates, influenced by the heterogeneity prior (τ), as illustrated by the sensitivity analyses using wide priors. Importantly, this difference does not relevantly affect the overall interpretation of the association between lead exposure and IQ. The two frameworks differ in how they treat outlier studies reporting substantial uncertainty. Bayesian meta-analysis can be less sensitive to these outliers, due to the shrinkage exerted by the prior specified for heterogeneity, provided that priors reflecting plausible domain knowledge are specified carefully. In contrast, frequentist models implicitly rely on flat or non-informative priors, which result in greater sensitivity to studies reporting large uncertainty [59].

Conducting a Bayesian meta-analysis can be more demanding than frequentist approaches, requiring greater statistical knowledge, explicit consideration of modelling assumptions, and typically more programming effort [60]. While these demands may initially appear as disadvantages, they can equally be understood as methodological safeguards: by requiring researchers to state assumptions explicitly rather than accepting software defaults, the Bayesian approach encourages deeper engagement with the data and statistical model.

A key distinction between the two frameworks is in how they handle parameter uncertainty. Frequentist models yield point estimates with confidence intervals, whereas Bayesian models produce full posterior distributions. This offers distinct practical advantages, particularly for risk assessment and policy-making. First, any credible interval can be derived from the posterior distribution to meet the needs of researchers, decision-makers, or other stakeholders. These

allow direct probability statements about parameter values, unlike confidence intervals which are often incorrectly interpreted as probability statements about parameters. Second, posterior distributions enable probabilistic comparisons across multiple health risks, such as calculating the probability that one risk exceeds another. When meaningful effect sizes or risk thresholds are defined, exceedance probabilities can be directly obtained from the posterior, providing an intuitive and policy-relevant summary of uncertainty. Posterior distributions together with the specified statistical model can be used to generate predictions for theoretical future studies, offering an additional way to characterize between-study heterogeneity. They also enable researchers to estimate the population-level health impact of an exposure under specified conditions. This is known as Environmental Burden of Disease (EBD). In EBD studies, health impacts are quantified using metrics such as DALYs, allowing comparisons across health determinants within and across populations, countries, and time periods. Such assessments integrate both population exposure and the exposure–response relationship [61]. The distributional results of the present study enable probabilistic EBD assessments via Monte Carlo simulations without requiring additional assumptions about the uncertainty distribution of the exposure–response function.

Our results are consistent with previous toxicological and epidemiological evidence [13,15,16,19] as well as prior meta-analyses on the effects of lead on IQ. Direct comparison of effect magnitudes is, however, limited by methodological differences: some studies estimated effects per exposure category [20,62,63], while others reported partial r [14] or standardized mean differences [64].

The RoB sensitivity analysis indicated that results are generally robust to individual-study bias. Notably, the study published by Crump KS, Van Landingham C, Bowers TS, Cahoy D, Chandalia JK. [25], which is a re-analysis of the landmark study by Lanphear BP, Hornung R, Khoury J, Yolton K, Baghurst P, Bellinger DC, et al. [16] received a high RoB rating in our assessment. Given that this re-analysis has informed multiple downstream analyses, including EBD assessments, its potential to introduce bias is a concern. The primary driver of the high RoB rating was the use of stepwise backward-elimination regression, a method that has been extensively criticized [52,65,66] and is central to the findings of Crump et al. This underscores the value of the updated evidence synthesis provided here.

Substantial heterogeneity was observed in this meta-analysis, likely driven by differences in study populations (size, location, age, and exposure range) and statistical methods, particularly the covariate adjustment sets used. Two studies reporting null effects were conducted in relatively low-exposed populations [43,49], where limited exposure variation reduces the detectability of an effect. Differences in study location may also contribute: IQ tests not adapted to the target population can result in underestimated scores, which may apply to the study by Menezes-Filho et al. [51]. Additionally, studies reporting larger uncertainty tended to report larger effects, suggesting a possible presence of publication bias (additional file 5g). The Bayesian approach was less sensitive to estimates from such studies, because the results of individual studies with large uncertainty were shrunk towards the pooled estimate, which is a particular advantage when pooling a small number of studies.

This study has several limitations that should be considered when interpreting the results. First, the included studies were heterogeneous with respect to covariate adjustment, exposure range,

IQ measurement instrument and cohort characteristics, meaning that the coefficients pooled in this study may not represent exactly the same estimand. This limits the generalizability of the pooled estimate and complicates its interpretation for specific populations.

Publication bias was assessed through visual inspection only; formal statistical testing, such as Egger’s test, would provide a more rigorous evaluation. The visual assessment suggested the presence of publication bias, which may have led to overestimation of the pooled effect.

Third, we excluded studies that did not report effect estimates on the logarithmic scale of blood lead levels. While this reduced the number of studies available for pooling, exclusion was necessary to ensure interpretability, as combining effect estimates expressed on different scales would render the pooled estimate uninterpretable.

Despite these limitations, this study also has several important strengths. By applying a parallel approach to both meta-analytic models — including the use of identical effect estimates, a multilevel random-effects model, and I^2 as a common measure of heterogeneity — we enable a direct comparison of the frequentist and Bayesian frameworks. To our knowledge, this is the first study to formally conduct such a comparison in the field of environmental epidemiology. Beyond this methodological contribution, we provide an updated evidence synthesis on the association between lead exposure and children’s IQ based on a systematic literature review. All data and code, including the code used to produce this manuscript, are publicly available, enabling other researchers to verify and build on our analyses. Finally, we adhered to PRISMA and BARG reporting guidelines, along with the use of the raRoB tool for risk of bias assessment, ensuring consistency and rigor in study selection, data extraction, study quality appraisal, and statistical analysis.

Conclusions

This study provides both an updated evidence synthesis of epidemiological studies examining the association between lead exposure and children’s IQ, and a methodological comparison of the Bayesian and conventional frequentist frameworks for meta-analysis. The resulting pooled effect estimates are consistent with previous evidence and once again underscore the importance of reducing environmental lead contamination to reduce the public health burden this substance continues to impose.

Comparing both frameworks showed that the Bayesian approach offers distinct advantages for risk assessment and policy-making. Especially the ability to incorporate prior knowledge, comprehensively quantify uncertainty, and produce probabilistic outputs that are more directly useable for decision-makers. While the Bayesian framework currently demands greater statistical knowledge and programming effort than the frequentist approach, these barriers are likely to diminish if the framework gains wider adoption and training becomes more accessible. We believe that the benefits gained outweigh the investments made and therefore want to encourage researchers to consider Bayesian methods.

List of Abbreviations

Abbrevia- tion	Meaning
BARG	Bayesian Analysis Reporting Guidelines
BLL	Blood lead level
CI	Confidence interval
CrI	Credible interval
DALY	Disability-adjusted life years
EBD	Environmental Burden of Disease
ERF	Exposure-response function
EU	European Union
IQ	Intelligence quotient
LLM	Large language model
PRISMA	Preferred Reporting Items for Systematic reviews and Meta-Analyses
RoB	Risk of bias
SD	Standard deviation
SE	Standard error
UI	Uncertainty interval

Supplementary Information

- Additional file 1; .docx; PRISMA 2020 checklist; Table containing all PRISMA 2020 items and the location where the items are reported
- Additional file 2; .xlsx; Search queries; Full search queries for the systematic review
- Additional file 3; .xlsx; Literature review summary; Summary of the literature review process, incl. details on exclusion reasons and the effect estimate harmonization
- Additional file 4; .pdf; Calculation of I^2 in the Bayesian Framework; Detailed rationale and description for the calculation of I^2 in the Bayesian meta-analysis
- Additional file 5; .pdf; Supplementary file 5; Diagnostic plots for the Bayesian model and funnel plot
- Additional file 6; .pdf; Model outputs; Full model output for frequentist and Bayesian meta-analysis

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

All data and code needed to reproduce the current study are publicly available on GitHub, https://github.com/PaulinaSell/lead-IQ_meta-analysis [38].

Competing interests

The authors declare no competing interests.

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Authors' contributions

PS: Conceptualization, data curation, formal analysis, methodology, writing – original draft, writing – review and editing. MS and PP: Data curation, writing – review and editing. SS: Methodology, writing – review and editing. DP: Supervision, writing – review and editing. All authors read and approved the final manuscript.

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